

Juan Antonio Monleón de la Lluvia

Nuclear Engineer - PhD Student

CONTACT

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EDUCATION

Double Master in Industrial Engineering and Nuclear Science and Technology

Polytechnic University of Madrid 2021-2023

Bachelor in Industrial Technology Engineering

Polytechnic University of Madrid 2017-2021

TECHNICAL SKILLS

Nuclear Codes: MCNP, Serpent,
OpenMC, NJOY, ADVANTG

Programming: Python, Git, VS Code,
Claude Code

Web / App Dev: React, TypeScript, Tauri,
FastAPI

Documentation: LaTeX, Microsoft 365

LANGUAGES

Spanish Native language

English C2 Level

ABOUT ME

I am a passionate professional dedicated to the nuclear sector, driven by a strong interest in research and innovation. As a proactive person, I constantly seek out new learning opportunities to expand my knowledge and skills. I enjoy applying my programming expertise to solve complex problems and enhance both my daily work and personal projects.

RESEARCH EXPERIENCE

PhD in Neutronics

ASNR (formerly IRSN), France

Nov 2023 - Present

Thesis: "Nuclear Data Uncertainty Quantification for PWR Pressure Vessel Shielding Calculations"

- Propagating nuclear-data uncertainties through Monte Carlo neutronic simulations of PWR vessel ageing via sensitivity analysis and Monte Carlo sampling, including the re-evaluation of ^{56}Fe elastic angular distributions and covariances from EXFOR experimental data
- Applying variance-reduction techniques (ADVANTG weight windows) to improve computational efficiency in deep-penetration shielding calculations
- Supervised a Master's student (Mar-Sep 2025) on nuclear-data uncertainty quantification in source-term calculations of PWRs

Research Internship

SCK CEN, Belgium

Sep 2022 - Jun 2023

- Completed Master's thesis titled "**Reducing Spectrum-Driven Uncertainties with Variance Reduction Techniques**"
- Conducted neutronic and deep shielding analyses using MCNP and ADVANTG for the MYRRHA Project

Master Thesis Research

UPM, Spain

Jun 2023 - Oct 2023

- Developed "**Enhancing the EXFOR nuclear data library with Machine Learning Techniques**"
- Implemented outlier detection methods for proton-induced reactions

Master Internship

ENUSA Advanced Industries, Spain

Oct 2021 - Apr 2022

AWARDS

Best Academic Report for Nuclear Master

Polytechnic University of Madrid (2022-2023)

Best Master Thesis Award

SCK CEN, Belgium (2023)

Best Poster Award

Journées des Thèses of IRSN (2024) -

[View Poster](#)

COMMUNITY

JEFF Community

Active member — Joint Evaluated Fission and Fusion File (OECD/NEA)

SINBAD Community

Member — Shielding Integral Benchmark Archive & Database (OECD/NEA)

- Designed and developed a Python program for the analysis and inspection of fuel assemblies and spent fuel

PUBLICATIONS

- J. A. Monleon de la Lluvia, M. Brovchenko, D. Rochman, E. Dumonteil, "Towards Efficient Nuclear Data Uncertainty Quantification in Radiation Shielding Calculations", *Nuclear Science and Engineering*, 2025. DOI: 10.1080/00295639.2025.2510048 · pre-print 2025

CONFERENCES & PRESENTATIONS

FIRST AUTHOR

- "Nuclear Data Uncertainty Quantification for PWR Pressure Vessel Shielding Calculations" — RPSD 2026, South Korea (*accepted*) Oct 2026
- "KIKa: A Unified Desktop Application for Nuclear Data Processing, Sensitivity, and Uncertainty Quantification" — RPSD 2026, South Korea (*accepted*) Oct 2026
- "Evaluation of elastic angular distribution covariances for ^{56}Fe from EXFOR experimental data" — WONDER 2026, Aix-en-Provence (*upcoming*) Jun 2026
- "Impact of Elastic Scattering Angular Distribution Covariances on Fast Neutron Fluence; Application to ^{56}Fe " — PHYSOR 2026 Apr 2026
- "Nuclear Data Uncertainty Quantification Using the VENUS-3 Benchmark Experiment" — ND 2025, Madrid 2025
- "Sensitivity Analysis and Uncertainty Quantification in PWR Irradiation Ageing-like problems" — ANS Winter Conference RPSD, Orlando Nov 2024

CO-AUTHORED

- Co-author: "Sensitivity Coefficients of a Single Response Function in Fixed-Source Calculations Using SERPENT2 and MCNP6" — PHYSOR 2026 Apr 2026
- Co-author: "Propagation of Angular Distribution Uncertainties in Vessel Fluence Calculations" — PHYSOR 2026 Apr 2026

SOFTWARE & TOOLS

KIKa — Nuclear-data Desktop Application

Lead developer 2025 - Present

- Cross-platform desktop app (Windows / macOS / Linux) for nuclear-data visualization, sensitivity analysis, and uncertainty quantification
- Tauri + React + TypeScript frontend with a FastAPI + Python backend; built on top of **kika-nd**, the open-source Python library (PyPI) providing the computational core (ENDF / ACE, covariances, EXFOR, MCNP interop)

TRAINING

- OpenMC Course — NEA, Paris Dec 2025
- Master in Management and Direction of SMEs | EFEM 2024-2025
- NJOY Course - NEA, Paris Dec 2024

- Sensitivity Analysis Summer School (SAMO) | UNIPR, Parma Jun 2024
- Serpent Bootcamp | NEA, Paris Nov 2023
- Course in "Nuclear Data for Energy and Non-Energy Applications" | GREAT PIONEER Dec 2022

REFERENCES

Dr. Mariya Brovchenko

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